

Studies in Real and Complex Analysis
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Metric Spaces and Continuity

The goal of topology is to study properties of geometric objects that are invariant under continuous deformation. The definition of a topological space encodes exactly the information needed to make this precise, but it seemingly comes out of nowhere without first developing some intuition. We therefore begin with a more concrete class of examples—metric spaces—where the notion of “closeness” is governed by a distance function, and see how the key ideas (open sets, continuity, convergence) arise naturally before abstracting to the general setting.

1.1 Metric Spaces

1.1.1 Definitions and Examples

Definition 1.1.1: Metric space

A metric space (X, d) is a set X together with a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$, called the distance function (or metric), satisfying:

- (i) $d(p, q) = 0 \iff p = q$ (identity of indiscernibles);
- (ii) $d(p, q) = d(q, p)$ (symmetry);
- (iii) $d(p, r) \leq d(p, q) + d(q, r)$ for all $p, q, r \in X$ (triangle inequality).

Example 1.1.1 (Euclidean space)

$X = \mathbb{R}^n$ with the Euclidean distance

$$d(x, y) = \left(\sum_{i=1}^n (y_i - x_i)^2 \right)^{1/2}.$$

For $n = 1$, this reduces to $d(a, b) = |a - b|$.

Example 1.1.2 (Alternative metrics on $\mathbb{R}^n$)

The following are also metrics on $\mathbb{R}^n$:

$$d_1(x, y) = \sum_{i=1}^n |y_i - x_i|, \quad d_\infty(x, y) = \max_{1 \leq i \leq n} |y_i - x_i|.$$

These are called the taxicab (or diamond) metric and the square (or sup) metric, respectively. The balls in these metrics are diamonds and squares rather than round disks.

Example 1.1.3 (Induced metric)

If (X, d) is a metric space and $Y \subseteq X$, then $(Y, d|_{Y \times Y})$ is a metric space with the restricted distance function. This is the induced metric (or subspace metric) on Y . For instance, any subset of $\mathbb{R}^n$ inherits the Euclidean metric.

Example 1.1.4 (The discrete metric)

On any set X , define $d(p, q) = 1$ if $p \neq q$ and $d(p, p) = 0$. This satisfies all three axioms (the triangle inequality holds since $d(p, r) = 1 \leq 1 + 1$ whenever $p \neq r$). In this metric, every subset is open and every point is isolated—the resulting topology is the discrete topology, which is a metric topology.

Example 1.1.5 (Function spaces)

Let X be any set and consider the space of bounded functions $f : X \rightarrow \mathbb{R}$. The sup metric is

$$d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

This is a metric: the triangle inequality follows from $|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|$ for each x , then taking the supremum. Convergence in this metric is uniform convergence: $f_n \rightarrow f$ iff $\sup_x |f_n(x) - f(x)| \rightarrow 0$.

Note:-

All three metrics d , d_1 , and d_∞ on $\mathbb{R}^n$ satisfy $d_\infty(x, y) \leq d(x, y) \leq d_1(x, y) \leq n \cdot d_\infty(x, y)$. In particular, they determine the same open sets (hence the same topology). We will see that different metrics can induce the same topology—the metric contains “extraneous information” beyond the topological structure.

1.1.2 Open Balls and Open Sets**Definition 1.1.2: Open ball**

Given a metric space (X, d) , a point $p \in X$, and $r > 0$, the open ball (or neighborhood) of radius r around p is

$$B_r(p) = \{q \in X : d(p, q) < r\}.$$

Definition 1.1.3: Open set

A subset $U \subseteq X$ is open if for every $p \in U$, there exists $\varepsilon > 0$ such that $B_\varepsilon(p) \subseteq U$. In words: U contains an open ball around each of its points.

Theorem 1.1.1 Open balls are open

Every open ball $B_r(p)$ is an open set.

Proof: Let $q \in B_r(p)$, so $d(p, q) = s < r$. Set $h = r - s > 0$. We claim $B_h(q) \subseteq B_r(p)$. For any $x \in B_h(q)$:

$$d(p, x) \leq d(p, q) + d(q, x) < s + h = r,$$

so $x \in B_r(p)$. Thus q is an interior point of $B_r(p)$. □

Proposition 1.1.1 Open sets form a topology

In any metric space X :

- (a) $\emptyset$ and X are open.
- (b) Any union of open sets is open.
- (c) Any finite intersection of open sets is open.

Proof: (a) $\emptyset$ is vacuously open. For X : every ball $B_\varepsilon(p) \subseteq X$.

(b) Let $\{U_\alpha\}$ be a collection of open sets and $p \in \bigcup_\alpha U_\alpha$. Then $p \in U_\beta$ for some β , so there exists $\varepsilon > 0$ with $B_\varepsilon(p) \subseteq U_\beta \subseteq \bigcup_\alpha U_\alpha$.

(c) Let $U_1, \dots, U_n$ be open and $p \in \bigcap_{i=1}^n U_i$. For each i , choose $\varepsilon_i > 0$ with $B_{\varepsilon_i}(p) \subseteq U_i$. Set $\varepsilon = \min(\varepsilon_1, \dots, \varepsilon_n) > 0$ (this is where finiteness is used). Then $B_\varepsilon(p) \subseteq U_i$ for all i , so $B_\varepsilon(p) \subseteq \bigcap_{i=1}^n U_i$. $\square$

Note:-

The finiteness condition in (c) is essential. For instance, $\bigcap_{n=1}^\infty (-1/n, 1/n) = \{0\}$, which is not open in $\mathbb{R}$. The infimum of infinitely many positive numbers can be zero.

Note:-

Every open set is a union of open balls. Given U open, for each $p \in U$ choose $\varepsilon_p > 0$ with $B_{\varepsilon_p}(p) \subseteq U$. Then $U = \bigcup_{p \in U} B_{\varepsilon_p}(p)$. So the open balls form a basis for the collection of open sets—they generate all open sets via unions.

1.1.3 Closed Sets

Definition 1.1.4: Closed set

A subset $Z \subseteq X$ is closed if its complement $X \setminus Z$ is open.

Note:-

A set can be both open and closed (e.g., $\emptyset$ and X itself), or neither open nor closed (e.g., $[0, 1]$ in $\mathbb{R}$). “Closed” does not mean “not open.”

Proposition 1.1.2 Properties of closed sets

By De Morgan’s laws applied to the open set axioms:

- (a) $\emptyset$ and X are closed.
- (b) Any intersection of closed sets is closed.
- (c) Any finite union of closed sets is closed.

Proof: These follow from complementation: $X \setminus \bigcap_\alpha Z_\alpha = \bigcup_\alpha (X \setminus Z_\alpha)$ and $X \setminus \bigcup_{i=1}^n Z_i = \bigcap_{i=1}^n (X \setminus Z_i)$. $\square$

Example 1.1.6 (Closed ball)

The closed ball $\overline{B}_r(p) = \{q \in X : d(p, q) \leq r\}$ is closed: its complement $\{q : d(p, q) > r\}$ is open (if $d(p, q) = s > r$, then $B_{s-r}(q)$ is contained in the complement by the triangle inequality).

Example 1.1.7 (Singletons are closed)

In any metric space, $\{p\}$ is closed. Indeed, $X \setminus \{p\} = \bigcup_{n=1}^\infty \{q : d(p, q) > 1/n\}$ is a union of open sets.

1.1.4 Limit Points, Closure, Interior, and Boundary

Definition 1.1.5: Limit point

A point $x \in X$ is a limit point (or accumulation point) of a subset $A \subseteq X$ if every open set U containing x satisfies $U \cap (A \setminus \{x\}) \neq \emptyset$. Equivalently, every ball $B_r(x)$ contains a point of A other than x itself.

Definition 1.1.6: Isolated point

A point $p \in A$ that is not a limit point of A is called an isolated point of A : there exists $r > 0$ with $B_r(p) \cap A = \{p\}$.

Theorem 1.1.2 Limit points yield infinitely many points

If p is a limit point of E , then every neighborhood of p contains infinitely many points of E .

Proof: Suppose some $B_r(p)$ contains only finitely many points $q_1, \dots, q_n$ of $E \setminus \{p\}$. Set $h = \min_i d(p, q_i) > 0$. Then $B_h(p) \cap (E \setminus \{p\}) = \emptyset$, contradicting the assumption that p is a limit point. $\square$

Corollary 1.1.1 Finite sets have no limit points

If A is finite, then A has no limit points. In particular, every finite subset of a metric space is closed.

Definition 1.1.7: Closure

The closure of $A \subseteq X$ is the smallest closed set containing A :

$$\bar{A} = \bigcap_{\substack{F \supseteq A \\ F \text{ closed}}} F.$$

This is closed (as an arbitrary intersection of closed sets). Equivalently, $x \in \bar{A}$ iff every open set containing x intersects A , i.e., $B_r(x) \cap A \neq \emptyset$ for all $r > 0$.

Proposition 1.1.3 Closure via limit points

For any $A \subseteq X$, $\bar{A} = A \cup A'$ where $A' = \{\text{limit points of } A\}$.

Proof: If $x \in A$, then every open set containing x intersects A (it contains $x \in A$), so $x \in \bar{A}$. If $x \in A'$, then every open set containing x meets $A \setminus \{x\} \subseteq A$, so $x \in \bar{A}$. Thus $A \cup A' \subseteq \bar{A}$.

Conversely, if $x \notin A \cup A'$, then $x \notin A$ and x is not a limit point of A , so there exists $r > 0$ with $B_r(x) \cap (A \setminus \{x\}) = \emptyset$. Since $x \notin A$, we get $B_r(x) \cap A = \emptyset$, so $x \notin \bar{A}$. $\square$

Corollary 1.1.2 Closed iff contains limit points

A is closed if and only if A contains all its limit points, i.e., $A' \subseteq A$. Equivalently, A is closed iff $\bar{A} = A$.

Definition 1.1.8: Interior

The interior of $A \subseteq X$ is the largest open set contained in A :

$$\text{Int}(A) = \bigcup_{\substack{U \subseteq A \\ U \text{ open}}} U.$$

A point $p \in \text{Int}(A)$ is called an interior point of A : there exists a neighborhood $B_r(p) \subseteq A$. We have A is open iff $\text{Int}(A) = A$.

Definition 1.1.9: Boundary

The boundary of A is $\partial A = \bar{A} \setminus \text{Int}(A)$. A point $x \in \partial A$ iff every open set containing x meets both A and $X \setminus A$.

Note:-

We also have the complementary identities $\overline{X \setminus A} = X \setminus \text{Int}(A)$ and $\text{Int}(X \setminus A) = X \setminus \bar{A}$.

Definition 1.1.10: Dense subset

A subset $A \subseteq X$ is dense in X if $\bar{A} = X$, i.e., every nonempty open subset of X intersects A .

Example 1.1.8 (Dense subsets of $\mathbb{R}$)

Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$ (with the usual topology). For $\mathbb{Q}$: given any open interval (a, b) , there exists a rational number in (a, b) by the Archimedean property.

Example 1.1.9 (Working in $\mathbb{C} \cong \mathbb{R}^2$)

Let all sets be subsets of $\mathbb{C}$ with the Euclidean metric $d(z, w) = |z - w|$.

- (a) $\{z \in \mathbb{C} : |z| < 1\}$: open (it is the ball $B_1(0)$), not closed (the sequence $1 - 1/n$ converges to $1 \notin A$), not perfect, bounded.
- (b) $\{z \in \mathbb{C} : |z| \leq 1\}$: not open (no ball around 1 fits inside), closed, perfect (every point is a limit point), bounded.
- (c) A nonempty finite set: not open, closed (finite sets have no limit points, so closure condition is vacuous), not perfect, bounded.

1.2 Continuity

1.2.1 The ε - δ Definition

Definition 1.2.1: Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \rightarrow Y$ is continuous at a point $p \in X$ if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x \in X, d_X(p, x) < \delta \implies d_Y(f(p), f(x)) < \varepsilon.$$

Equivalently, $f(B_\delta(p)) \subseteq B_\varepsilon(f(p))$. We say f is continuous if it is continuous at every point of X .

1.2.2 The Open Preimage Characterization

Theorem 1.2.1 Continuity via open sets

Let $f : X \rightarrow Y$ be a map between metric spaces. Then f is continuous if and only if the preimage $f^{-1}(U)$ of every open set $U \subseteq Y$ is open in X .

Proof: ($\implies$) Assume f is continuous, and let $U \subseteq Y$ be open. Take any $p \in f^{-1}(U)$, so $f(p) \in U$. Since U is open, there exists $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq U$. By continuity at p , there exists $\delta > 0$ with $f(B_\delta(p)) \subseteq B_\varepsilon(f(p)) \subseteq U$. So $B_\delta(p) \subseteq f^{-1}(U)$, and $f^{-1}(U)$ is open.

($\Leftarrow$) Assume $f^{-1}(U)$ is open for every open $U \subseteq Y$. Fix $p \in X$ and $\varepsilon > 0$. The ball $B_\varepsilon(f(p))$ is open in Y , so $f^{-1}(B_\varepsilon(f(p)))$ is open in X . Since $p \in f^{-1}(B_\varepsilon(f(p)))$, there exists $\delta > 0$ with $B_\delta(p) \subseteq f^{-1}(B_\varepsilon(f(p)))$. This means $d_X(p, x) < \delta \implies f(x) \in B_\varepsilon(f(p))$, i.e., $d_Y(f(p), f(x)) < \varepsilon$. $\square$

Note:-

This theorem is the bridge from metric spaces to topology. It says continuity depends only on which sets are open—not on the specific values of the distance function. This motivates defining a topological space as a set equipped with a collection of “open sets” satisfying the axioms of Proposition 1, and defining continuity as: $f^{-1}(\text{open}) = \text{open}$.

1.3 Sequences and Limits

1.3.1 Convergence

Definition 1.3.1: Convergent sequence

A sequence $p_1, p_2, p_3, \dots$ in a metric space (X, d) converges to a limit $p \in X$, written $p_n \rightarrow p$ or $\lim_{n \rightarrow \infty} p_n = p$, if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, d(p_n, p) < \varepsilon.$$

If no such p exists, the sequence diverges.

Theorem 1.3.1 Properties of convergent sequences

Let (X, d) be a metric space.

- (a) $\{p_n\}$ converges to p if and only if every open set $U \ni p$ contains p_n for all but finitely many n .
- (b) Limits are unique: if $p_n \rightarrow a$ and $p_n \rightarrow b$, then $a = b$.
- (c) If $\{p_n\}$ converges, then $\{p_n\}$ is bounded (i.e., the set $\{p_n : n \geq 1\}$ is contained in some ball).
- (d) If p is a limit point of $E \subseteq X$, then there is a sequence $\{p_n\}$ in E with $p_n \neq p$ for all n and $p_n \rightarrow p$.

Proof: (a) ($\Rightarrow$) If $U \ni p$ is open, then $B_\varepsilon(p) \subseteq U$ for some $\varepsilon > 0$, so $p_n \in B_\varepsilon(p) \subseteq U$ for $n \geq N$. ($\Leftarrow$) For any $\varepsilon > 0$, the ball $B_\varepsilon(p)$ is open and contains p , so it contains p_n for all but finitely many n . Thus $d(p_n, p) < \varepsilon$ for large n .

(b) Let $\varepsilon > 0$. Choose N_1, N_2 so that $d(p_n, a) < \varepsilon/2$ for $n \geq N_1$ and $d(p_n, b) < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$: $d(a, b) \leq d(a, p_n) + d(p_n, b) < \varepsilon$. Since ε is arbitrary, $d(a, b) = 0$, so $a = b$.

(c) Choose N so that $d(p_n, p) < 1$ for $n \geq N$. Set $r = \max(1, d(p_1, p), \dots, d(p_N, p))$. Then $d(p_n, p) \leq r$ for all n .

(d) For each positive integer n , there exists $p_n \in E$ with $d(p_n, p) < 1/n$ and $p_n \neq p$ (since p is a limit point). Given $\varepsilon > 0$, choose N with $N > 1/\varepsilon$. Then $n > N$ implies $d(p_n, p) < 1/n < \varepsilon$, so $p_n \rightarrow p$. $\square$

Proposition 1.3.1 Closed sets via sequences

A set $Z \subseteq X$ is closed if and only if whenever $\{p_n\}$ is a sequence in Z converging to some $p \in X$, we have $p \in Z$. That is, closed sets “contain the limits of their convergent sequences.”

Proof: ($\Rightarrow$) Suppose Z is closed and $p_n \in Z$ with $p_n \rightarrow p$. If $p \notin Z$, then $U = X \setminus Z$ is open and contains p . By part (a) of the theorem, U contains p_n for all but finitely many n —but every $p_n \in Z$, so $p_n \notin U$. Contradiction.

($\Leftarrow$) We show Z contains all its limit points. If p is a limit point of Z , then by part (d) there exists a sequence $p_n \in Z$ with $p_n \rightarrow p$, so $p \in Z$ by hypothesis. $\square$

1.3.2 Cauchy Sequences and Completeness

Definition 1.3.2: Cauchy sequence

A sequence $\{p_n\}$ in (X, d) is a Cauchy sequence if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N, d(p_m, p_n) < \varepsilon.$$

Proposition 1.3.2 Convergent implies Cauchy

Every convergent sequence is Cauchy.

Proof: If $p_n \rightarrow p$, then for any $\varepsilon > 0$, choose N with $d(p_n, p) < \varepsilon/2$ for $n \geq N$. For $m, n \geq N$: $d(p_m, p_n) \leq d(p_m, p) + d(p, p_n) < \varepsilon$. $\square$

Note:-

The converse fails in general. In $\mathbb{Q}$ (with the usual metric), the sequence of partial sums $x_n = \sum_{k=0}^n 1/k!$ is Cauchy, but its limit e is irrational, so $\{x_n\}$ has no limit in $\mathbb{Q}$. The notion of Cauchy sequence captures “wants to converge” without reference to a limit.

Definition 1.3.3: Complete metric space

A metric space (X, d) is complete if every Cauchy sequence in X converges (to a point of X).

Example 1.3.1 (Complete and incomplete spaces)

$\mathbb{R}$ (with the usual metric) is complete—this is equivalent to the least upper bound property. More generally, $\mathbb{R}^n$ is complete. However, $\mathbb{Q}$ with the induced metric is not complete.

1.3.3 Interlude: What is $\mathbb{R}$?

The real numbers $\mathbb{R}$ are characterized (up to isomorphism) as the unique ordered field with the least upper bound property: every nonempty subset $E \subseteq \mathbb{R}$ that is bounded above has a least upper bound (supremum) $\sup(E) \in \mathbb{R}$. That is, $\sup(E)$ is an upper bound for E , and every upper bound u for E satisfies $u \geq \sup(E)$.

The l.u.b. property is equivalent to completeness of $\mathbb{R}$: the constructions of $\mathbb{R}$ from $\mathbb{Q}$ (Dedekind cuts or Cauchy sequence completion) both produce the same object. In the Cauchy completion approach, elements of $\mathbb{R}$ are equivalence classes of Cauchy sequences in $\mathbb{Q}$, where two sequences are equivalent if their difference converges to zero. This makes $\mathbb{R}$ the completion of $\mathbb{Q}$ —the smallest complete metric space containing $\mathbb{Q}$ as a dense subset.

Theorem 1.3.2 $\sup E$ belongs to $\bar{E}$

Let E be a nonempty subset of $\mathbb{R}$ that is bounded above, and let $y = \sup E$. Then $y \in \bar{E}$. In particular, if E is closed, then $\sup E \in E$.

Proof: For any $\varepsilon > 0$, $y - \varepsilon$ is not an upper bound for E (since y is the least upper bound), so there exists $x \in E$ with $y - \varepsilon < x \leq y$. Thus $B_\varepsilon(y) \cap E \neq \emptyset$, and $y \in \bar{E}$. $\square$

1.4 Limits of Functions

Definition 1.4.1: Limit of a function at a point

Let (X, d_X) and (Y, d_Y) be metric spaces, $E \subseteq X$, $f : E \rightarrow Y$, and p a limit point of E . We write $\lim_{x \rightarrow p} f(x) = q$ if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x \in E, 0 < d_X(x, p) < \delta \implies d_Y(f(x), q) < \varepsilon.$$

Note the condition $0 < d_X(x, p)$: we exclude the point p itself (which may or may not belong to E).

Theorem 1.4.1 Sequential characterization of limits

Let f, E, p be as above. Then $\lim_{x \rightarrow p} f(x) = q$ if and only if $\lim_{n \rightarrow \infty} f(p_n) = q$ for every sequence $\{p_n\}$ in E with $p_n \neq p$ and $p_n \rightarrow p$.

Corollary 1.4.1 Uniqueness of function limits

If the limit $\lim_{x \rightarrow p} f(x)$ exists, it is unique.

Proof of the corollary: If $f(p_n) \rightarrow q$ and $f(p_n) \rightarrow q'$ for some sequence $p_n \rightarrow p$, then $q = q'$ by uniqueness of sequence limits. $\square$

Note:-

Continuity of f at $p \in E$ (when p is a limit point of E) is equivalent to $\lim_{x \rightarrow p} f(x) = f(p)$. When p is an isolated point of E , f is automatically continuous at p (the ε - δ condition is satisfied by taking δ small enough that $B_\delta(p) \cap E = \{p\}$).

1.5 Further Properties of Metric Spaces

1.5.1 Bounded Sets and Diameter

Definition 1.5.1: Bounded set

A subset E of a metric space (X, d) is bounded if there exist a point $q \in X$ and a real number $M > 0$ such that $d(p, q) < M$ for all $p \in E$. Equivalently, E is contained in some ball.

Definition 1.5.2: Diameter

The diameter of a nonempty subset $E \subseteq X$ is

$$\text{diam}(E) = \sup\{d(p, q) : p, q \in E\}.$$

Then E is bounded iff $\text{diam}(E) < \infty$.

Note:-

Boundedness is not a topological property: it depends on the specific metric, not just on which sets are open. For instance, $f : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$ defined by $f(x) = \tan(x)$ is a homeomorphism (both f and $f^{-1} = \arctan$ are continuous), yet the domain is bounded and the range is not.

1.5.2 Open Sets Relative to a Subspace

Theorem 1.5.1 Relative openness

Suppose $Y \subseteq X$ is a subspace with the induced metric, and $E \subseteq Y$. Then E is open relative to Y if and only if $E = Y \cap G$ for some open set $G \subseteq X$.

Proof: ($\Leftarrow$) If G is open in X and $p \in E = Y \cap G$, then there exists $\varepsilon > 0$ with $B_\varepsilon^X(p) \subseteq G$. So $B_\varepsilon^Y(p) = B_\varepsilon^X(p) \cap Y \subseteq G \cap Y = E$, and E is open in Y .

($\Rightarrow$) If E is open in Y , then for each $p \in E$ choose $\varepsilon_p > 0$ with $B_{\varepsilon_p}^Y(p) \subseteq E$. Set $G = \bigcup_{p \in E} B_{\varepsilon_p}^X(p)$. Then G is open in X and $G \cap Y = \bigcup_{p \in E} (B_{\varepsilon_p}^X(p) \cap Y) = \bigcup_{p \in E} B_{\varepsilon_p}^Y(p) = E$. $\square$

Note:-

This is the definition of the subspace topology: the open sets in Y are exactly $\{U \cap Y : U \text{ open in } X\}$. Beware: a set can be open in Y without being open in X . For example, $(0, 1) \cap \mathbb{Q}$ is open in $\mathbb{Q}$ (with the induced metric from $\mathbb{R}$) but not open in $\mathbb{R}$.

1.6 More on Continuity

1.6.1 Sequential Characterization

Proposition 1.6.1 Continuity and sequences

Let $f : X \rightarrow Y$ be a map between metric spaces. Then f is continuous at $p \in X$ if and only if $p_n \rightarrow p$ in X implies $f(p_n) \rightarrow f(p)$ in Y .

Proof: ($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ with $f(B_\delta(p)) \subseteq B_\varepsilon(f(p))$. Since $p_n \rightarrow p$, there exists N with $d_X(p_n, p) < \delta$ for $n \geq N$. Then $d_Y(f(p_n), f(p)) < \varepsilon$ for $n \geq N$.

($\Leftarrow$) Suppose f is not continuous at p . Then there exists $\varepsilon > 0$ such that for every $\delta > 0$, there is a point x with $d_X(x, p) < \delta$ but $d_Y(f(x), f(p)) \geq \varepsilon$. Taking $\delta = 1/n$, we get a sequence $p_n \rightarrow p$ with $d_Y(f(p_n), f(p)) \geq \varepsilon$ for all n , contradicting $f(p_n) \rightarrow f(p)$. $\square$

Proposition 1.6.2 Compositions of continuous maps

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Proof: For any open $W \subseteq Z$, $g^{-1}(W)$ is open in Y (since g is continuous), so $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$ is open in X (since f is continuous). $\square$

1.6.2 Uniform Continuity

Definition 1.6.1: Uniform continuity

A map $f : X \rightarrow Y$ between metric spaces is uniformly continuous if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall p, q \in X, d_X(p, q) < \delta \implies d_Y(f(p), f(q)) < \varepsilon.$$

The difference from ordinary continuity: δ depends only on ε , not on the point p .

Example 1.6.1 (Uniformly continuous vs. continuous)

The function $f(x) = x^2$ on $(\mathbb{R}, |\cdot|)$ is continuous but not uniformly continuous: for $\varepsilon = 1$, no single δ works for all p (near p , we need $\delta \approx 1/(2|p|)$, which shrinks as $|p| \rightarrow \infty$).

The function $f(x) = \sqrt{x}$ on $[0, \infty)$ is uniformly continuous.

Definition 1.6.2: Lipschitz map

A map $f : X \rightarrow Y$ is Lipschitz (with constant L) if $d_Y(f(p), f(q)) \leq L \cdot d_X(p, q)$ for all $p, q \in X$. Every Lipschitz map is uniformly continuous (take $\delta = \varepsilon/L$).

Example 1.6.2

The distance function $d(p, \cdot) : X \rightarrow \mathbb{R}$ is Lipschitz with constant 1: $|d(p, x) - d(p, y)| \leq d(x, y)$ by the triangle inequality. In particular, $d(p, \cdot)$ is continuous.

1.7 The Contraction Mapping Theorem

A contraction is a Lipschitz map with constant strictly less than 1. The following is one of the most useful theorems in analysis.

Theorem 1.7.1 Banach Fixed-Point Theorem

Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction: there exists $0 \leq c < 1$ such that $d(T(x), T(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Then T has a unique fixed point $x^* \in X$ (i.e., $T(x^*) = x^*$). Moreover, for any starting point $x_0 \in X$, the iterates $x_n = T(x_{n-1}) = T^n(x_0)$ converge to x^* .

Proof: Uniqueness. If $T(x^*) = x^*$ and $T(y^*) = y^*$, then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq c \cdot d(x^*, y^*)$. Since $c < 1$, this forces $d(x^*, y^*) = 0$, so $x^* = y^*$.

Existence. Fix any $x_0 \in X$ and define $x_n = T(x_{n-1})$. We show $\{x_n\}$ is Cauchy. First, $d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq c \cdot d(x_n, x_{n-1}) \leq \dots \leq c^n \cdot d(x_1, x_0)$. For $m > n$:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq d(x_1, x_0) \sum_{k=n}^{m-1} c^k \leq \frac{c^n}{1-c} \cdot d(x_1, x_0).$$

Since $c^n \rightarrow 0$, the sequence is Cauchy. By completeness, $x_n \rightarrow x^*$ for some $x^* \in X$.

Fixed point. Since T is continuous (it is Lipschitz), $T(x^*) = T(\lim x_n) = \lim T(x_n) = \lim x_{n+1} = x^*$. $\square$

Note:-

The contraction mapping theorem is used throughout analysis: to prove the inverse function theorem, the existence and uniqueness theorem for ODEs (Picard–Lindelöf), and the implicit function theorem. The key ingredients are completeness (to guarantee the limit exists) and the contraction (to guarantee convergence).

1.8 Exercises

Exercise 1.8.1 Metric spaces and continuity exercises

- (1) Verify that d_1 and d_∞ on $\mathbb{R}^n$ satisfy the axioms of a metric. Show that $d_\infty(x, y) \leq d(x, y) \leq \sqrt{n} \cdot d_\infty(x, y)$ where d is the Euclidean metric.
- (2) Show that a finite intersection of open sets is open directly from the ε - δ definition (without

appealing to the proposition). Give an example showing that an infinite intersection of open sets need not be open.

- (3) Let $X = \{0\} \cup \{1/n : n \in \mathbb{Z}_{>0}\} \subset \mathbb{R}$ with the induced metric $d(x, y) = |x - y|$. Prove directly (without using completeness of $\mathbb{R}$) that X is complete.
- (4) Let $f : X \rightarrow Y$ be a continuous map between metric spaces. Prove that f maps convergent sequences to convergent sequences: if $p_n \rightarrow p$ in X , then $f(p_n) \rightarrow f(p)$ in Y .
- (5) Let $f, g : X \rightarrow Y$ be continuous maps between metric spaces, and suppose Y is Hausdorff (in particular, any metric space). Show that $\{x \in X : f(x) = g(x)\}$ is closed in X .
- (6) Prove that a map $f : X \rightarrow Y$ between metric spaces is continuous if and only if $f^{-1}(Z)$ is closed in X for every closed set $Z \subseteq Y$.
- (7) Describe the open balls in $(\mathbb{R}^2, d_1)$ and $(\mathbb{R}^2, d_\infty)$. Verify that the three metrics d, d_1, d_∞ on $\mathbb{R}^2$ induce the same topology: a set is open in one metric if and only if it is open in the others.
- (8) Let $T : [0, 1] \rightarrow [0, 1]$ be defined by $T(x) = \frac{1}{3}x + \frac{1}{2}$. Show that T is a contraction, find its fixed point, and verify that the iterates $T^n(x_0)$ converge to it for any $x_0 \in [0, 1]$.
- (9) Prove that if $f : X \rightarrow Y$ is uniformly continuous and $\{x_n\}$ is a Cauchy sequence in X , then $\{f(x_n)\}$ is a Cauchy sequence in Y . Show by example that this can fail if f is merely continuous.

Solution

(1) For d_∞ : positivity and symmetry are clear from $|y_i - x_i| \geq 0$ and $|y_i - x_i| = |x_i - y_i|$. Identity: $d_\infty(x, y) = 0$ iff $\max_i |y_i - x_i| = 0$ iff $|y_i - x_i| = 0$ for all i iff $x = y$. Triangle inequality: for each i , $|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i| \leq d_\infty(x, y) + d_\infty(y, z)$, so $d_\infty(x, z) = \max_i |x_i - z_i| \leq d_\infty(x, y) + d_\infty(y, z)$. The proof for d_1 is similar (sum the scalar triangle inequalities).

For the comparison: $d_\infty(x, y) = \max_i |y_i - x_i| \leq \sqrt{\sum_i |y_i - x_i|^2} = d(x, y)$ since the max is at most the ℓ^2 -norm. And $d(x, y)^2 = \sum_i |y_i - x_i|^2 \leq n \cdot \max_i |y_i - x_i|^2 = n \cdot d_\infty(x, y)^2$, giving $d(x, y) \leq \sqrt{n} \cdot d_\infty(x, y)$.

(2) Let $U_1, \dots, U_n$ be open and $p \in \bigcap U_i$. For each i , choose $\varepsilon_i > 0$ with $B_{\varepsilon_i}(p) \subseteq U_i$. Then $\varepsilon = \min(\varepsilon_1, \dots, \varepsilon_n) > 0$ satisfies $B_\varepsilon(p) \subseteq U_i$ for all i . Counterexample: $\bigcap_{n=1}^\infty (-1/n, 1/n) = \{0\}$, which is not open in $\mathbb{R}$.

(3) Let $\{x_n\}$ be a Cauchy sequence in X . If $x_n = 0$ for infinitely many n , then since the sequence is Cauchy, eventually all terms are close to 0, forcing $x_n \rightarrow 0 \in X$. Otherwise, $x_n \neq 0$ for all $n \geq N_0$, so $x_n = 1/k_n$ for positive integers k_n . The Cauchy condition with $\varepsilon < \min_{m \neq n} |1/k_m - 1/k_n|$ eventually forces the sequence to be constant: $x_n = 1/k$ for all $n \geq N$, and $1/k \in X$.

More precisely: for $\varepsilon = 1/2$, there exists N such that $|x_m - x_n| < 1/2$ for $m, n \geq N$. If $x_N = 0$, then $|x_n| < 1/2$ for $n \geq N$, so $x_n \in \{0\} \cup \{1\}$. Since $|x_n - 0| < 1/2$, we get $x_n = 0$ for all $n \geq N$. If $x_N = 1/k$ for some $k \geq 2$, then $|x_n - 1/k| < 1/2$, and since the elements of X near $1/k$ are discrete (spaced at least $1/k - 1/(k+1) = 1/(k(k+1))$ apart), the sequence is eventually constant.

(4) Let $p_n \rightarrow p$ in X and $\varepsilon > 0$. By continuity of f at p , there exists $\delta > 0$ with $f(B_\delta(p)) \subseteq B_\varepsilon(f(p))$. Since $p_n \rightarrow p$, there exists N with $d_X(p_n, p) < \delta$ for $n \geq N$. Then $d_Y(f(p_n), f(p)) < \varepsilon$ for $n \geq N$.

(5) Let $A = \{x \in X : f(x) = g(x)\}$. We show $X \setminus A$ is open. If $x \notin A$, then $f(x) \neq g(x)$. Since Y is a metric space (hence Hausdorff), there exist $\varepsilon > 0$ with $B_\varepsilon(f(x)) \cap B_\varepsilon(g(x)) = \emptyset$ (take $\varepsilon = d_Y(f(x), g(x))/2$). By continuity, choose $\delta > 0$ with $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$ and $g(B_\delta(x)) \subseteq B_\varepsilon(g(x))$. For any $y \in B_\delta(x)$: $f(y) \in B_\varepsilon(f(x))$ and $g(y) \in B_\varepsilon(g(x))$, so $f(y) \neq g(y)$ (the balls are disjoint), hence $y \notin A$. Thus $B_\delta(x) \subseteq X \setminus A$.

(6) ($\Rightarrow$) If f is continuous and $Z \subseteq Y$ is closed, then $Y \setminus Z$ is open, so $f^{-1}(Y \setminus Z) = X \setminus f^{-1}(Z)$ is open, meaning $f^{-1}(Z)$ is closed. ($\Leftarrow$) If $U \subseteq Y$ is open, then $Y \setminus U$ is closed, so $f^{-1}(Y \setminus U) = X \setminus f^{-1}(U)$ is closed, meaning $f^{-1}(U)$ is open.

(7) In $(\mathbb{R}^2, d_1)$, the ball $B_r(p)$ is a square rotated 45° (a diamond): $\{(x, y) : |x - p_1| + |y - p_2| < r\}$. In $(\mathbb{R}^2, d_\infty)$, the ball is an axis-aligned square: $\{(x, y) : |x - p_1| < r \text{ and } |y - p_2| < r\}$. To show equivalent topologies: if U is open in d , then for any $p \in U$ there exists $\varepsilon > 0$ with $B_\varepsilon^d(p) \subseteq U$. Since

$B_\varepsilon^{d_\infty}(\mathbf{p}) \subseteq B_\varepsilon^{\mathbf{d}}(\mathbf{p})$ (because $d_\infty \leq d$), the set U is also d_∞ -open. Conversely, $B_\varepsilon^{\mathbf{d}}(\mathbf{p}) \subseteq B_{\frac{\varepsilon}{\sqrt{2}}}^{d_\infty}(\mathbf{p})$ gives the reverse inclusion. The argument for d_1 is analogous using $d_\infty \leq d \leq d_1 \leq 2d_\infty$.

(8) $T(x) = x/3 + 1/2$ satisfies $|T(x) - T(y)| = |x - y|/3$, so it is a contraction with $c = 1/3$. The fixed point solves $x = x/3 + 1/2$, giving $2x/3 = 1/2$, so $x^* = 3/4$. For any $x_0 \in [0, 1]$:

$$|T^n(x_0) - 3/4| = (1/3)^n |x_0 - 3/4| \leq (1/3)^n \rightarrow 0.$$

So $T^n(x_0) \rightarrow 3/4$ regardless of starting point.

(9) Let $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ with $d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$. Since $\{x_n\}$ is Cauchy, choose N with $d_X(x_m, x_n) < \delta$ for $m, n \geq N$. Then $d_Y(f(x_m), f(x_n)) < \varepsilon$ for $m, n \geq N$, so $\{f(x_n)\}$ is Cauchy.

Counterexample: $f(x) = 1/x$ on $(0, 1)$ is continuous. The sequence $x_n = 1/n$ is Cauchy in $(0, 1)$, but $f(x_n) = n$ is not Cauchy (not even bounded). The function f is not uniformly continuous on $(0, 1)$.

Topological Spaces

In **Lecture 1** we saw that continuity between metric spaces can be characterized purely in terms of open sets (Theorem 3.4 of the Harris handout): f is continuous iff $f^{-1}(\text{open}) = \text{open}$. This means the metric itself is extraneous for the purposes of continuity, limits, and connectedness—all that matters is which sets are declared open. This motivates the following abstraction.

2.1 The Definition

Definition 2.1.1: Topology

A topology on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X , called the open sets, satisfying:

- (i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- (ii) The union of any subcollection of elements of $\mathcal{T}$ is in $\mathcal{T}$.
- (iii) The intersection of any finite subcollection of elements of $\mathcal{T}$ is in $\mathcal{T}$.

A topological space is a pair $(X, \mathcal{T})$. When the topology is clear, we just write X . A subset $U \subseteq X$ is open if $U \in \mathcal{T}$.

Example 2.1.1 (The metric topology)

If (X, d) is a metric space, the collection $\mathcal{T} = \{U \subseteq X : U \text{ is open in the metric sense}\}$ is a topology on X (proved in Lecture 1). This is the metric topology induced by d .

Example 2.1.2 (The discrete and indiscrete topologies)

On any set X : the discrete topology is $\mathcal{T} = \mathcal{P}(X)$ (every subset is open). This is the metric topology for the discrete metric $d(x, y) = 1$ if $x \neq y$. The indiscrete (or trivial) topology is $\mathcal{T} = \{\emptyset, X\}$. In the indiscrete topology, the only open sets are the empty set and X itself.

Example 2.1.3 (The finite complement topology)

Let X be any set. Define $\mathcal{T}_f = \{U \subseteq X : X \setminus U \text{ is finite}\} \cup \{\emptyset\}$. This is a topology: $\emptyset$ and X are in $\mathcal{T}_f$ (since $X \setminus X = \emptyset$ is finite). If $\{U_\alpha\}$ is a collection of elements of $\mathcal{T}_f$, then $X \setminus \bigcup U_\alpha = \bigcap (X \setminus U_\alpha)$, which is an intersection of finite sets, hence finite. If $U_1, \dots, U_n \in \mathcal{T}_f$, then $X \setminus \bigcap U_i = \bigcup (X \setminus U_i)$, a finite union of finite sets, hence finite.

Example 2.1.4 (Topologies on a three-element set)

Let $X = \{a, b, c\}$. Examples of topologies include $\{\emptyset, X\}$ (indiscrete), $\{\emptyset, \{a\}, X\}$, $\{\emptyset, \{a, b\}, \{c\}, X\}$, $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{b, c\}, X\}$, and $\mathcal{P}(X)$ (discrete). The collection $\{\emptyset, \{a\}, \{b\}, X\}$ is not a topology: it is missing $\{a\} \cup \{b\} = \{a, b\}$.

Note:-

Why bother with topologies that don't come from a metric? In analysis, many natural topologies are not metrizable. The space of bounded functions $f : X \rightarrow \mathbb{R}$ with the topology of pointwise convergence ($f_n \rightarrow f$ iff $f_n(x) \rightarrow f(x)$ for all x) does not come from a metric in general (it is a product topology). The C^∞ topology on smooth functions $\mathbb{R} \rightarrow \mathbb{R}$ is also not metrizable. On the other hand, a metric contains extraneous information: the metrics d, d_1, d_∞ on $\mathbb{R}^n$ all induce the same topology.

2.2 Continuous Maps Between Topological Spaces

Definition 2.2.1: Continuous map

A map $f : X \rightarrow Y$ between topological spaces is continuous if $f^{-1}(U)$ is open in X for every open set $U \subseteq Y$.

Definition 2.2.2: Convergence in topological spaces

A sequence $\{p_n\}$ in a topological space X converges to $p \in X$, written $p_n \rightarrow p$, if for every open set $U \ni p$, there exists N such that $p_n \in U$ for all $n \geq N$. Equivalently, every open set containing p contains all but finitely many p_n .

Note:-

In a general topological space, limits of sequences need not be unique. In the indiscrete topology $\mathcal{T} = \{\emptyset, X\}$, every sequence converges to every point (the only open set containing p is X itself). We will need the Hausdorff condition to guarantee uniqueness.

Proposition 2.2.1 Basic properties of continuous maps

- (a) If $f : X \rightarrow Y$ is continuous and $p_n \rightarrow p$ in X , then $f(p_n) \rightarrow f(p)$ in Y .
- (b) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Proof: (a) Let $V \ni f(p)$ be open in Y . Then $f^{-1}(V)$ is open in X and contains p , so it contains p_n for all but finitely many n . Thus $f(p_n) \in V$ for all but finitely many n .

(b) If $W \subseteq Z$ is open, then $g^{-1}(W)$ is open in Y , so $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$ is open in X . □

2.2.1 Comparing Topologies

Definition 2.2.3: Finer and coarser

Given two topologies $\mathcal{T}$ and $\mathcal{T}'$ on X , we say $\mathcal{T}'$ is finer than $\mathcal{T}$ (or $\mathcal{T}$ is coarser than $\mathcal{T}'$) if $\mathcal{T} \subseteq \mathcal{T}'$. If $\mathcal{T} \subsetneq \mathcal{T}'$, we say $\mathcal{T}'$ is strictly finer. Two topologies are comparable if one contains the other.

Note:-

The discrete topology is the finest topology on X (every point is isolated), and the indiscrete topology is the coarsest. A finer topology has more open sets: it is easier for functions into X to be continuous (more preimages are open), but harder for sequences to converge (more open sets to check). In the discrete topology, a sequence converges iff it is eventually constant. In the indiscrete topology, every sequence converges to every point.

2.3 Bases for a Topology

Keeping track of all open sets is cumbersome. In metric spaces, we started with open balls and generated all open sets as unions. The abstract analogue is a basis.

Definition 2.3.1: Basis for a topology

A basis for a topology on X is a collection $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfying:

- (i) For every $x \in X$, there exists $B \in \mathcal{B}$ with $x \in B$.
- (ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

The topology $\mathcal{T}$ generated by $\mathcal{B}$ consists of all sets that are unions of elements of $\mathcal{B}$. Equivalently, $U \in \mathcal{T}$ iff for every $x \in U$, there exists $B \in \mathcal{B}$ with $x \in B \subseteq U$.

Note:-

Condition (i) ensures $X \in \mathcal{T}$ (since $X = \bigcup_{B \in \mathcal{B}} B$). Condition (ii) ensures finite intersections of open sets are open: if $U_1, U_2 \in \mathcal{T}$ and $x \in U_1 \cap U_2$, pick $B_1, B_2 \in \mathcal{B}$ with $x \in B_i \subseteq U_i$, then (ii) gives $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$, so $U_1 \cap U_2 \in \mathcal{T}$. Unlike bases in linear algebra, topological bases can contain redundant elements—a better analogy is with generating sets.

Example 2.3.1 (Bases for familiar topologies)

- The open balls $\{B_r(x) : x \in X, r > 0\}$ form a basis for the metric topology on X . Even the rational-radius balls $\{B_{1/n}(x) : x \in X, n \geq 1\}$ suffice.
- The open intervals $\{(a, b) : a < b, a, b \in \mathbb{R}\}$ form a basis for the usual (standard) topology on $\mathbb{R}$. Condition (ii): $(a_1, b_1) \cap (a_2, b_2) = (\max(a_1, a_2), \min(b_1, b_2))$ is either empty or another open interval.

Definition 2.3.2: The lower limit topology

$\mathbb{R}_\ell$ is $\mathbb{R}$ with the topology generated by the basis of half-open intervals $\{[a, b) : a < b\}$. This is strictly finer than the standard topology: every open interval $(a, b) = \bigcup_n [a + 1/n, b)$ is open in $\mathbb{R}_\ell$, but $[0, 1)$ is open in $\mathbb{R}_\ell$ and not in the standard topology.

Definition 2.3.3: The K-topology

$\mathbb{R}_K$ is $\mathbb{R}$ with the topology generated by the basis $\{(a, b) : a < b\} \cup \{(a, b) \setminus K : a < b\}$ where $K = \{1/n : n \in \mathbb{Z}_{>0}\}$. This is strictly finer than the standard topology but is not comparable to $\mathbb{R}_\ell$.

Proposition 2.3.1 Continuity via bases

Let $\mathcal{B}_X$ and $\mathcal{B}_Y$ be bases for topologies on X and Y . Then $f : X \rightarrow Y$ is continuous iff for every $B \in \mathcal{B}_Y$ and every $x \in f^{-1}(B)$, there exists $B' \in \mathcal{B}_X$ with $x \in B' \subseteq f^{-1}(B)$. In metric spaces, this reduces to the familiar ε - δ condition (with $B = B_\varepsilon(f(x))$ and $B' = B_\delta(x)$).

2.4 Subspace, Product, and Quotient Topologies

2.4.1 The Subspace Topology

Definition 2.4.1: Subspace topology

Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$. The subspace topology on Y is $\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}$. That is, a set is open in Y iff it is the intersection of Y with an open set of X .

Note:-

When stating “ U is open,” it is important to specify in which space. The set $(0, 1)$ is open in $\mathbb{R}$, open in $[0, 2]$ (since $(0, 1) = (-1, 1) \cap [0, 2]$), but not open in $\mathbb{R}^2$. The set Y is always open in itself (as $Y = X \cap Y$). If the metric on X restricts to Y , the subspace topology agrees with the induced metric topology: if $\mathcal{T}_X$ comes from a metric d on X , then $\mathcal{T}_Y$ comes from $d|_{Y \times Y}$.

Note:-

The subspace topology is the coarsest topology on Y making the inclusion $\iota : Y \hookrightarrow X$ continuous.

2.4.2 The Product Topology

Definition 2.4.2: Product topology (finite case)

Given topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$, the product topology on $X \times Y$ is the topology generated by the basis

$$\mathcal{B} = \{U \times V : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}.$$

This is a basis: condition (i) holds since $X \times Y \in \mathcal{B}$, and condition (ii) since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2) \in \mathcal{B}$.

Note:-

When X, Y are metric spaces, the product topology on $X \times Y$ is the metric topology for $d_\infty((x_1, y_1), (x_2, y_2)) = \max(d_X(x_1, x_2), d_Y(y_1, y_2))$. This agrees with the product topologies for the metrics $d_2 = \sqrt{d_X^2 + d_Y^2}$ and $d_1 = d_X + d_Y$, since all three induce the same topology. The key observation is that $B_r^{d_\infty}((x, y)) = B_r^X(x) \times B_r^Y(y)$, so the metric balls are exactly the basis elements.

Proposition 2.4.1 Continuity into products

A map $f : Z \rightarrow X \times Y$ is continuous iff both components $f_1 = \pi_X \circ f : Z \rightarrow X$ and $f_2 = \pi_Y \circ f : Z \rightarrow Y$ are continuous, where π_X, π_Y are the projection maps. Equivalently, a sequence $(x_n, y_n) \rightarrow (x, y)$ in $X \times Y$ iff $x_n \rightarrow x$ in X and $y_n \rightarrow y$ in Y .

Proof: The projections π_X, π_Y are continuous (preimage of $U \subseteq X$ open is $U \times Y$, which is open). So if f is continuous, then $f_1 = \pi_X \circ f$ and $f_2 = \pi_Y \circ f$ are compositions of continuous maps. Conversely, if f_1, f_2 are continuous and $U \times V$ is a basis element of $X \times Y$, then $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$, a finite intersection of open sets, hence open. Since every open set in $X \times Y$ is a union of such basis elements, f^{-1} of any open set is open. $\square$

2.4.3 Infinite Products: Box vs. Product Topology

For a family of spaces $\{(X_i, \mathcal{T}_i)\}_{i \in I}$, there are two natural topologies on $\prod_{i \in I} X_i$.

Definition 2.4.3: Box topology

The box topology on $\prod_{i \in I} X_i$ has basis consisting of all products $\prod_{i \in I} U_i$ where each $U_i \subseteq X_i$ is open.

Definition 2.4.4: Product topology

The product topology on $\prod_{i \in I} X_i$ has basis consisting of all products $\prod_{i \in I} U_i$ where each $U_i \subseteq X_i$ is open and $U_i = X_i$ for all but finitely many i .

Note:-

For finite I , the two topologies coincide. For infinite I , the product topology is strictly coarser than the box topology. The product topology is the “right” choice: it is the coarsest topology making all projections $\pi_i : \prod X_j \rightarrow X_i$ continuous. The characterization of continuity extends: $f : Z \rightarrow \prod X_i$ is continuous (in the product topology) iff each component $f_i = \pi_i \circ f$ is continuous.

Example 2.4.1 (The diagonal map)

Let $\Delta : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$ be defined by $\Delta(x) = (x, x, x, \dots)$. In the product topology, Δ is continuous (each component $\pi_n \circ \Delta = \text{id}_{\mathbb{R}}$ is continuous). In the box topology, Δ is not continuous: the set $U = \prod_{n=1}^{\infty} (-1/n, 1/n)$ is open in the box topology, but $\Delta^{-1}(U) = \bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$, which is not open in $\mathbb{R}$.

2.4.4 Homeomorphisms

Definition 2.4.5: Homeomorphism

Two topological spaces X and Y are homeomorphic, written $X \cong Y$, if there exist continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. Equivalently, f is a bijection such that both f and f^{-1} are continuous. Such an f is called a homeomorphism. A homeomorphism is an isomorphism of topological spaces: it induces a bijection $\mathcal{T}_X \leftrightarrow \mathcal{T}_Y$.

Note:-

A continuous bijection need not be a homeomorphism: the inverse may fail to be continuous. If $\mathcal{T}'$ is strictly finer than $\mathcal{T}$ on X , then $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is a continuous bijection (since $\mathcal{T} \subseteq \mathcal{T}'$), but $\text{id}^{-1} : (X, \mathcal{T}) \rightarrow (X, \mathcal{T}')$ is not continuous (some $U \in \mathcal{T}' \setminus \mathcal{T}$ fails to pull back to an open set).

Example 2.4.2

$f : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$ defined by $f(x) = \tan(x)$ is a homeomorphism (both $\tan$ and $\arctan$ are continuous). So $\mathbb{R}$ is homeomorphic to any bounded open interval. This shows boundedness is not a topological property.

2.5 Closed Sets, Closure, Interior, Boundary

All the definitions from Lecture 1 carry over verbatim to general topological spaces.

Definition 2.5.1: Closed set

A subset A of a topological space X is closed if $X \setminus A$ is open. By De Morgan: $\emptyset$ and X are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

Note:-

A set can be both open and closed (clopen): $\emptyset$ and X always are, and in the discrete topology every set is clopen. A set can also be neither open nor closed: $[0, 1]$ in the standard topology on $\mathbb{R}$, or $\mathbb{Q} \subset \mathbb{R}$.

Definition 2.5.2: Closure, interior, boundary

Let $A \subseteq X$ be any subset.

- The closure of A is $\bar{A} = \bigcap \{F \supseteq A : F \text{ closed}\}$, the smallest closed set containing A . A point $x \in \bar{A}$ iff every open set containing x meets A .
- The interior of A is $\text{Int}(A) = \bigcup \{U \subseteq A : U \text{ open}\}$, the largest open set contained in A . A point $x \in \text{Int}(A)$ iff some open set containing x is contained in A .
- The boundary of A is $\partial A = \bar{A} \setminus \text{Int}(A)$. A point $x \in \partial A$ iff every open set containing x meets both $\bar{A}$ and $X \setminus A$.

Note:-

The complementary identities hold: $\overline{X \setminus A} = X \setminus \text{Int}(A)$ and $\text{Int}(X \setminus A) = X \setminus \bar{A}$. Also: A is closed iff $\bar{A} = A$, and A is open iff $\text{Int}(A) = A$.

Example 2.5.1

In $\mathbb{R}$ with the standard topology: if $A = [0, 1]$, then $\bar{A} = [0, 1]$, $\text{Int}(A) = (0, 1)$, and $\partial A = \{0, 1\}$.

Definition 2.5.3: Neighborhood

An open set U containing a point p is called a neighborhood of p . Then: $p \in \text{Int}(A)$ iff A contains a neighborhood of p , and $p \in \bar{A}$ iff every neighborhood of p intersects A .

Definition 2.5.4: Dense subset

A is dense in X if $\bar{A} = X$, i.e., every nonempty open set intersects A . For instance, $\mathbb{Q}$ is dense in $\mathbb{R}$ (standard topology).

2.6 Limit Points and the Hausdorff Condition

2.6.1 Limit Points

Definition 2.6.1: Limit point

A point $x \in X$ is a limit point of $A \subseteq X$ if every neighborhood U of x satisfies $U \cap (A \setminus \{x\}) \neq \emptyset$. The set of limit points of A is denoted A' . A point of A that is not a limit point is an isolated point of A .

Proposition 2.6.1 Closure via limit points

$\bar{A} = A \cup A'$. In particular, A is closed iff $A' \subseteq A$ (iff A contains all its limit points).

Proof: If $x \in A$, every neighborhood meets A (it contains x). If $x \in A'$, every neighborhood meets $A \setminus \{x\} \subseteq A$. Either way $x \in \bar{A}$. Conversely, if $x \notin A$ and $x \notin A'$, then some neighborhood U satisfies $U \cap (A \setminus \{x\}) = \emptyset$; since $x \notin A$, this gives $U \cap A = \emptyset$, so $x \notin \bar{A}$. $\square$

Example 2.6.1

In $\mathbb{R}_{\text{std}}$: 1 is a limit point of both $(0, 1)$ and $[0, 1]$. In the set $S = \{1/n : n \geq 1\} \cup \{0\}$, the point 0 is a limit point (every neighborhood of 0 contains $1/n$ for large n), but 1 is an isolated point (the ball $B_{1/2}(1) \cap S = \{1\}$).

Note:-

In metric spaces, $x \in \overline{A}$ iff there exists a sequence in A converging to x , and x is a limit point of A iff there exists a sequence in $A \setminus \{x\}$ converging to x . In general topological spaces, sequences are insufficient: the correct tool is nets or filters. However, in spaces with a countable basis of neighborhoods at each point (“first-countable” spaces, which include all metric spaces), sequences suffice.

2.6.2 Hausdorff Spaces

In the indiscrete topology on $\mathbb{R}$, every sequence converges to every point, and in the finite complement topology on $\mathbb{R}$, a sequence of distinct points converges to every point of $\mathbb{R}$. To exclude such pathologies:

Definition 2.6.2: Hausdorff space

A topological space X is Hausdorff (or T_2) if for every pair of distinct points $x_1 \neq x_2 \in X$, there exist disjoint open sets U_1, U_2 with $x_1 \in U_1$ and $x_2 \in U_2$.

Theorem 2.6.1 Hausdorff implies unique limits

If X is Hausdorff, then every sequence in X converges to at most one limit.

Proof: Suppose $x_n \rightarrow x$ and $y \neq x$. Choose disjoint open sets $U_x \ni x$ and $U_y \ni y$. Since $x_n \rightarrow x$, there exists N with $x_n \in U_x$ for all $n \geq N$. But then $x_n \notin U_y$ for $n \geq N$ (since $U_x \cap U_y = \emptyset$), so $\{x_n\}$ does not converge to y . $\square$

Example 2.6.2 (Hausdorff and non-Hausdorff spaces)

- (a) Every metric space is Hausdorff: if $x \neq y$, set $r = d(x, y)/2 > 0$ and take $U = B_r(x)$, $V = B_r(y)$.
- (b) The finite complement topology on $\mathbb{R}$ is not Hausdorff: any two nonempty open sets have finite complements, so their intersection is $\mathbb{R}$ minus a finite set, hence nonempty. But it is T_1 (singletons are closed, since $\mathbb{R} \setminus \{x\}$ has finite complement $\{x\}$).
- (c) The discrete topology is always Hausdorff ($U_i = \{x_i\}$).
- (d) Subspaces of Hausdorff spaces are Hausdorff. Products of Hausdorff spaces are Hausdorff.

Note:-

There is a hierarchy of separation axioms: T_1 (points are closed) $\Leftarrow T_2$ (Hausdorff) $\Leftarrow T_3$ (regular) $\Leftarrow T_4$ (normal). The finite complement topology on $\mathbb{R}$ is T_1 but not T_2 . We will primarily work with Hausdorff spaces. Every metric space is T_4 (hence satisfies all separation axioms).

2.7 Connectedness

Definition 2.7.1: Connected space

A topological space X is connected if it cannot be written as $X = U \cup V$ where U, V are disjoint, nonempty, open sets. Such a decomposition is called a separation of X . A subset $A \subseteq X$ is connected if it is connected in the subspace topology.

Proposition 2.7.1 $[0, 1]$ is connected

The interval $[0, 1] \subset \mathbb{R}$ (with the standard topology) is connected.

Proof: Suppose $[0, 1] = U \cup V$ is a separation with $0 \in U$. Let $a = \sup\{x \in [0, 1] : [0, x] \subseteq U\}$. Since U is open and $0 \in U$, we have $a > 0$.

If $a \in V$: since V is open in $[0, 1]$, there exists $\varepsilon > 0$ with $(a - \varepsilon, a] \cap [0, 1] \subseteq V$. But by definition of a , the interval $[0, a - \varepsilon)$ is not entirely in U , so some point of $[0, a - \varepsilon)$ is in V . But then there exist points of V below $a - \varepsilon$, and $[0, x] \subseteq U$ fails before $a - \varepsilon$, contradicting $a = \sup$.

More carefully: if $a < 1$ and $a \in U$, then U is open, so $(a - \varepsilon, a + \varepsilon) \subseteq U$ for some ε , contradicting $a = \sup$. If $a \in V$, then $(a - \varepsilon, a + \varepsilon) \cap [0, 1] \subseteq V$, so $[0, x] \subseteq U$ for $x < a - \varepsilon$ only, giving $\sup \leq a - \varepsilon < a$, contradiction. So $a = 1$, meaning $U = [0, 1]$ and $V = \emptyset$. $\square$

Theorem 2.7.1 Continuous image of a connected space is connected

If $f : X \rightarrow Y$ is continuous and X is connected, then $f(X) \subseteq Y$ is connected.

Proof: Suppose $f(X) = U \cup V$ is a separation (in the subspace topology). Then $f^{-1}(U)$ and $f^{-1}(V)$ are disjoint, nonempty (since $U, V \subseteq f(X)$), and open in X (since f is continuous and U, V are open in $f(X)$, which means $U = U' \cap f(X)$ for some open $U' \subseteq Y$, so $f^{-1}(U) = f^{-1}(U')$ is open). Their union is X , giving a separation of X . Contradiction. $\square$

Corollary 2.7.1 Intermediate Value Theorem

Let X be a connected topological space and $f : X \rightarrow \mathbb{R}$ continuous. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $r \in f(X)$: there exists $c \in X$ with $f(c) = r$.

Proof: The image $f(X)$ is a connected subset of $\mathbb{R}$. If $r \notin f(X)$, then $U = f(X) \cap (-\infty, r)$ and $V = f(X) \cap (r, \infty)$ give a separation of $f(X)$ (both nonempty since $f(a)$ is on one side and $f(b)$ on the other). Contradiction. $\square$

Theorem 2.7.2 Connected subsets of $\mathbb{R}$ are intervals

A subset $A \subseteq \mathbb{R}$ is connected (in the standard topology) if and only if A is an interval (i.e., $x, z \in A$ and $x < y < z$ implies $y \in A$).

2.7.1 Path-Connectedness

Definition 2.7.2: Path

A path in X from x to y is a continuous map $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. (Here $[0, 1]$ has the subspace topology from $\mathbb{R}$.)

Definition 2.7.3: Path-connected

A space X is path-connected if for every pair $x, y \in X$, there exists a path from x to y . The relation $x \sim y$ (“connected by a path”) is an equivalence relation; the equivalence classes are the path components of X .

Theorem 2.7.3 Path-connected implies connected

If X is path-connected, then X is connected.

Proof: Suppose $X = U \cup V$ is a separation. Pick $x \in U, y \in V$, and let $\gamma : [0, 1] \rightarrow X$ be a path from x to y . Then $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are disjoint open sets covering $[0, 1]$, with $0 \in \gamma^{-1}(U)$ and $1 \in \gamma^{-1}(V)$, giving a separation of $[0, 1]$. This contradicts the connectedness of $[0, 1]$. $\square$

Note:-

The converse is false in general: the topologist’s sine curve $S = \{(x, \sin(1/x)) : x > 0\} \cup \{(0, y) : -1 \leq y \leq 1\}$ is connected but not path-connected. However, for open subsets of $\mathbb{R}^n$, connected and path-connected are equivalent.

Theorem 2.7.4 Products of connected spaces are connected

If X_i is connected for each $i \in I$, then $\prod_{i \in I} X_i$ (with the product topology) is connected.

2.8 Exercises

Exercise 2.8.1 Topology exercises

- (1) Verify that the finite complement topology $\mathcal{T}_f$ on an infinite set X is T_1 but not Hausdorff.
- (2) Let X be a set with topologies $\mathcal{T}$ and $\mathcal{T}'$, with $\mathcal{T}'$ strictly finer than $\mathcal{T}$. Show that $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is continuous but not a homeomorphism.
- (3) Show that if X is Hausdorff, then $Y \subseteq X$ (with the subspace topology) is Hausdorff, and $X \times Y$ (with the product topology) is Hausdorff whenever both factors are.
- (4) Prove that $A \subseteq \mathbb{R}$ (standard topology) is connected iff A is an interval.
- (5) Let X be a topological space and $\{A_i\}_{i \in I}$ a family of connected subspaces with $\bigcap_{i \in I} A_i \neq \emptyset$. Show that $\bigcup_{i \in I} A_i$ is connected.
- (6) Show that the closure of a connected set is connected.
- (7) Prove that a finite topological space is metrizable if and only if the topology is discrete.

Solution

- (1) T_1 : for any $x \in X$, $X \setminus \{x\}$ has complement $\{x\}$, which is finite, so $X \setminus \{x\} \in \mathcal{T}_f$. Thus singletons are closed. Not Hausdorff: if U, V are nonempty open sets, then $X \setminus U$ and $X \setminus V$ are finite, so $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite. Since X is infinite, $U \cap V$ is infinite (in particular nonempty). So no two nonempty open sets are disjoint.
- (2) Continuity: if $U \in \mathcal{T}$, then $U \in \mathcal{T}'$ (since $\mathcal{T} \subseteq \mathcal{T}'$), so $\text{id}^{-1}(U) = U$ is open in $\mathcal{T}'$. Not a homeomorphism: since $\mathcal{T}' \supsetneq \mathcal{T}$, there exists $V \in \mathcal{T}' \setminus \mathcal{T}$. Then $(\text{id}^{-1})^{-1}(V) = V \notin \mathcal{T}$, so id^{-1} is not continuous.

- (3) Subspace: if $y_1 \neq y_2 \in Y \subseteq X$, take disjoint open $U_1, U_2 \subseteq X$ separating them. Then $U_1 \cap Y, U_2 \cap Y$ are disjoint open sets in Y separating y_1, y_2 . Product: given $(x_1, y_1) \neq (x_2, y_2)$ in $X \times Y$, WLOG $x_1 \neq x_2$. Choose disjoint open $U_1, U_2 \subseteq X$ with $x_i \in U_i$. Then $U_1 \times Y$ and $U_2 \times Y$ are disjoint open sets in $X \times Y$ separating the two points.
- (4) ($\Rightarrow$) If A is not an interval, there exist $x < y < z$ with $x, z \in A$ and $y \notin A$. Then $A = (A \cap (-\infty, y)) \cup (A \cap (y, \infty))$ is a separation. ($\Leftarrow$) If A is an interval and $A = U \cup V$ is a separation with $u \in U, v \in V$, WLOG $u < v$. Let $a = \sup(U \cap [u, v])$. If $a \in U$: U open gives $(a - \varepsilon, a + \varepsilon) \cap A \subseteq U$, contradicting $a = \sup$ (there are points of U beyond a). If $a \in V$: V open gives $(a - \varepsilon, a + \varepsilon) \cap A \subseteq V$, so $\sup(U \cap [u, v]) \leq a - \varepsilon < a$, contradiction.
- (5) Suppose $Y = \bigcup A_i = U \cup V$ is a separation. Pick $p \in \bigcap A_i$. WLOG $p \in U$. For each i , the sets $U \cap A_i$ and $V \cap A_i$ are disjoint open sets in A_i covering A_i , with $p \in U \cap A_i \neq \emptyset$. Since A_i is connected, $V \cap A_i = \emptyset$ for all i . Thus $V \cap Y = \bigcup (V \cap A_i) = \emptyset$, contradicting $V \neq \emptyset$.
- (6) Let A be connected and suppose $\bar{A} = U \cup V$ is a separation. Then $A = (U \cap A) \cup (V \cap A)$ with $U \cap A$ and $V \cap A$ open in A and disjoint. Since A is connected, one is empty, say $V \cap A = \emptyset$, so $A \subseteq U$. But then $\bar{A} \subseteq \bar{U}$. Since U is open in $\bar{A}$ and V is its complement in $\bar{A}$, U is also closed in $\bar{A}$, giving $\bar{U} \cap \bar{A} = U$. So $\bar{A} \subseteq U$, meaning $V = \bar{A} \setminus U = \emptyset$, contradiction.
- (7) ($\Leftarrow$) The discrete topology on a finite set $X = \{x_1, \dots, x_n\}$ is metrizable (use the discrete metric). ($\Rightarrow$) Suppose X is finite and metrizable with metric d . For each $x \in X$, let $r_x = \min\{d(x, y) : y \in X, y \neq x\} > 0$ (the minimum exists since X is finite). Then $B_{r_x}(x) = \{x\}$, so singletons are open. Since every subset is a union of singletons, every subset is open, and the topology is discrete.